

The Matching Polytope

$$G = (V, E)$$

$$Q_{PM} = \left\{ x \geq 0, \quad x(\delta(v)) = 1, \quad x(E[u]) \leq 1, \quad \forall u \in V, \quad |u| \text{ odd} \right\} \quad (*)$$

Last lecture: $Q_{PM} = P_{PM}$

$$Q_M = \left\{ x \geq 0, \quad x(\delta(v)) \leq 1, \quad x(E[u]) \leq \left\lfloor \frac{1}{2} |u| \right\rfloor, \quad u \in V, \quad |u| \text{ odd} \right\}$$

This lecture: $Q_M = P_M \stackrel{\Delta}{=} \text{conv} \left\{ x^M : M \text{ a matching} \right\}$

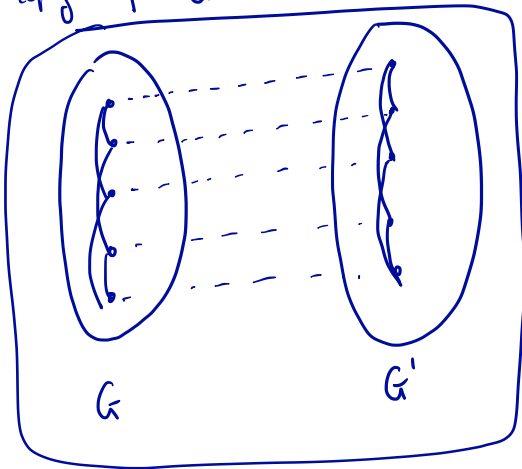
Corollary: $Q_{PM} = P_{PM} = \left\{ x \geq 0, \quad x(\delta(v)) = 1, \quad x(E[u]) \leq \left\lfloor \frac{1}{2} |u| \right\rfloor, \quad u \in V, \quad |u| \text{ odd} \right\}$

Proof — 1/3

As usual, $P_M \subseteq Q_M$.

For the reverse inclusion:

Make a copy of G : $G' = (V', E')$



$$\tilde{G} = (\tilde{V}, \tilde{E})$$

$$\tilde{V} = V \cup V'$$

$$\tilde{E} = E \cup E' \cup \{(v, v'), v, v' \in V\}$$

Proof — 2/3

Take any $x \in Q_M$.

$x \in \mathbb{R}^E$. Extend x to $\tilde{x} \in \mathbb{R}^{\tilde{E}}$ as follows:

$$\tilde{x}_e = \tilde{x}_{e'} = x_e \quad \forall e \in E$$

$$\tilde{x}(e_{v,v'}) = \underline{1 - x(\delta(v))}, \quad \forall v \in V$$

Lemma: $\tilde{x} \in Q_{PM}(\tilde{G})$.

Corollary: $x \in P_M(G)$, hence concluding the proof.

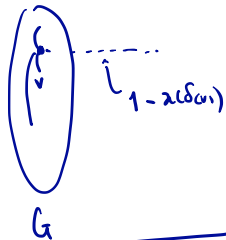
Exercise: Prove the corollary using the Lemma.

Proof — 3/3

$$\tilde{x} \geq 0. \checkmark$$

$$\tilde{x}(\tilde{\delta}(v)) = 1. \text{ Holds by construction.}$$

For any $v \in V$, since $x \in Q_M \Rightarrow \underline{x(\delta(v))} \leq 1$

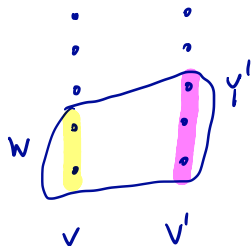


Have left to show that

$$\tilde{x} \text{ satisfies: } \tilde{x}(\tilde{\delta}(\tilde{u})) \geq 1,$$

$$\forall \tilde{u} \subseteq V \cup V', |\tilde{u}| \text{ odd.}$$

$$\text{Say: } \tilde{u} = W \cup Y'$$



Exercise :

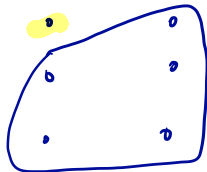
$$\tilde{x}(\tilde{\delta}(\tilde{u})) \geq \tilde{x}(\tilde{\delta}(W \setminus Y)) + \tilde{x}(\tilde{\delta}(Y' \setminus W'))$$

$$\tilde{u} = W u Y'$$

$$\begin{array}{cc} V & V' \\ \vdots & \vdots \end{array}$$

$|\tilde{u}|$ odd

$$|W \setminus Y| \approx |Y \setminus W|$$



must be odd. wlog: $|W \setminus Y|$ odd.

$$\star \quad \underline{\tilde{\kappa}(E[W \setminus Y])} \leq \left\lfloor \frac{1}{2} |W \setminus Y| \right\rfloor = \frac{1}{2} |W \setminus Y| - \frac{1}{2} \quad // \text{ b/c } \kappa \in \mathbb{Q}_M.$$

$$\underline{\tilde{\kappa}(\tilde{\delta}(W \setminus Y))} = \sum_{v \in W \setminus Y} \tilde{\kappa}(\tilde{\delta}(v)) - \underline{2\tilde{\kappa}(E[W \setminus Y])}$$

$$\geq |W \setminus Y| - |W \setminus Y| + 1 = \underline{1}.$$

$$\Rightarrow \tilde{\kappa}(\tilde{\delta}(\tilde{u})) \geq 1 \Rightarrow \tilde{\kappa} \in \mathcal{P}_{PM}(\tilde{u}). \Rightarrow (\text{by Corollary}) \Rightarrow \kappa \in \mathcal{P}_M(k)$$